

C# Lab - Day 08

Part 1: The Domain Model (`Student` Class)

- **Properties:**

- `Id` (int)
- `Name` (string)
- `Age` (int)
- `Gender` (string)
- `DepartmentName` (string)

- **Constructors:** Provide a parameterized constructor to initialize all the properties.

- **Methods:** Override the `ToString()` method to return a cleanly formatted string containing the student's details.

Part 2: The Data Source (`Repository` Class)

- **Methods:** Implement a method named `GetStudents()`.

- **Requirements:** This method must return a `List<Student>` pre-populated with varied student records to act as a mock DB.

Part 3: The Filtering Logic (`Filtration` Class)

- **Custom Delegate:** Define a custom delegate that points to a function taking a `Student` object as a parameter and returning a `bool`.
- **Methods:** Implement a method named `FilterByAny()`.
 - **Parameters:** It should accept a `List<Student>` and an instance of your custom delegate.
 - **Behavior:** Iterate through the list, apply the delegate condition, and return a filtered `List<Student>`.

Part 4: Execution (`Main` Method)

- **Data Initialization:** Call `GetStudents()` from the `Repository` class.
- **Test Custom Delegate:**
 - Create matching functions to be used with your delegate.
 - Use your `FilterByAny()` method along with your custom delegate to filter the students.
 - **Important:** You must implement an array instead of a list collection within the main program execution to manage and iterate over the final results before printing them to the console.
- **Test Built-in Delegates:**
 - Try built-in delegates.